# IoT in Real Life: Applications in Industrial Automation

## Introduction

The Internet of Things (IoT) is a network of physical devices fitted with sensors, processors and communication interfaces that collect and exchange data with little human intervention. In industry, this idea is usually called the Industrial Internet of Things (IIoT) and is a core pillar of Industry 4.0. Traditional automation relied on programmable logic controllers (PLCs) and SCADA systems isolated inside a single plant. IoT extends automation by linking machines, products, people and enterprise software into one connected, data-driven system (Xu, He and Li, 2014; Sisinni et al., 2018).

## Core Components of Industrial IoT

An industrial IoT system generally has four layers. The first is the sensing and actuation layer, made up of sensors that measure vibration, temperature, pressure, flow, position and energy use, together with actuators such as valves, drives and robotic arms. The second is the connectivity layer, which moves data through industrial protocols such as OPC UA, Modbus and MQTT and through wired Ethernet, Wi-Fi, 5G or low-power wireless networks. The third is the data processing layer, in which edge gateways filter and analyse data close to the machine for fast response, while cloud platforms store large datasets and run heavier analytics. The fourth is the application layer, which presents dashboards, alerts and control functions to engineers and managers (Boyes et al., 2018).

## Key Applications

**Predictive maintenance.** This is the most widely discussed IoT application in industry. Sensors attached to motors, pumps, bearings and conveyors continuously record vibration, temperature and current. Machine learning models compare this stream with normal behaviour and flag early signs of wear, so that repairs are scheduled before a breakdown occurs. Unlike fixed-interval maintenance, this approach reduces unplanned downtime and avoids replacing parts that are still healthy (Krupitzer et al., 2020). Rúbio, Dionísio and Torres (2019) demonstrated a smart device that classifies vibration faults in electric motors directly on the machine, which shows the practicality of edge-level analysis.

**Real-time monitoring and process control.** IoT lets operators observe production lines remotely and in real time. Data on output, cycle time, quality and energy consumption can be viewed from a central dashboard, and control parameters can be adjusted automatically when a process drifts from its target. This supports consistent product quality and quicker response to faults (Visconti et al., 2024).

**Asset tracking and supply chain management.** RFID tags, barcodes and GPS trackers record the location and condition of raw materials, work-in-progress and finished goods. Factories can see inventory levels immediately, reduce search time, and trigger automatic reordering.

**Robotics and smart production lines.** Connected robots and automated guided vehicles exchange data with each other and with production planning software. This allows flexible

manufacturing, where a line can switch between product variants quickly. Collaborative robots use sensor data to work safely beside human workers.

**Energy management.** Smart meters and sensors measure electricity, water and compressed air use at machine level. Analysis of this data identifies waste, such as idle equipment or leaks, and supports efficient scheduling of energy-intensive processes. (Visconti et al., 2024).

**Worker safety.** Wearable devices and environmental sensors detect hazards such as toxic gases, excessive noise, extreme heat or a worker entering a restricted zone near operating machinery. Equipment can be stopped automatically.

## Real-World Example

A typical case is a car manufacturing plant. Vibration and temperature sensors on assembly-line motors send readings to an edge gateway, which detects abnormal patterns and warns the maintenance team several days before a likely failure. RFID tags track each vehicle body through the paint shop and assembly stations, while robots receive the correct instructions for each model. The importance of such reliability is clear: Krupitzer et al. (2020) note that production problems at Volkswagen in 2016 caused sales losses of up to 400 million euros per week, which illustrates how costly unplanned stoppages can be for large manufacturers.

## Benefits

The main benefits of IoT in industrial automation are reduced downtime, lower maintenance and energy costs, better product quality, higher productivity and improved safety. Data collected from machines also supports better planning and evidence-based decisions. Remote monitoring also lets specialists support several sites without travelling.

## Challenges

Several challenges slow adoption. Cybersecurity is the most serious, since connecting operational equipment to networks creates new entry points for attackers, and an attack can halt production or endanger safety. Interoperability is another problem because plants often contain old machines and devices from many vendors using different protocols. Data volume and quality must be managed, as poorly calibrated sensors or incomplete data weaken analytical models. Initial investment and the need for skilled staff in data analytics and networking can be barriers, particularly for small and medium-sized enterprises (Sisinni et al., 2018).

## Conclusion

IoT has become a foundation of modern industrial automation. By connecting sensors, machines and software, it enables predictive maintenance, real-time control, efficient use of energy and materials, and safer workplaces. Although security, compatibility and cost remain obstacles, the technology offers clear gains in efficiency and reliability, and its role in factories will keep growing as analytics and connectivity improve.

## References

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